Ephedrine

Authors

Alec K. Statler; Christopher V. Maani[1]; Arpan Kohli[2].

Affiliations

[1] US Army
[2] West Virginia University

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Continuing Education Activity

Ephedrine is a medication used in the management and treatment of clinically significant hypotension. It is in the sympathomimetic class of medications. This activity illustrates the indications, action, and contraindications for ephedrine as a valuable agent in the treatment of clinically significant hypotension. This activity will highlight the mechanism of actions, adverse event profile, and other key factors pertinent to interprofessional healthcare team members involved in the care of patients with clinically significant hypotension and related conditions.

Objectives:

- Identify the indications for treatment with ephedrine.
- Describe the common complications of managing clinically significant hypotension intraoperatively with the ephedrine.
- Explain the mechanism of action and alternative options to treat significant intraoperative hypotension.
- Review the importance of communication among the interprofessional team to improve outcomes for the patient requiring ephedrine.

Access free multiple choice questions on this topic.

Indications

The FDA-approved primary indication for ephedrine is the treatment of clinically significant hypotension perioperatively. Induction of general anesthesia and ongoing anesthesia during operative cases results in vasodilatation and hypotension requiring treatment with vasopressors.[1] It is also frequently the agent of choice for hypotension induced by spinal or epidural anesthesia. In obstetrics, sympathectomy during spinal anesthetics results in hypotension in 80% of patients.[2] Although phenylephrine selection is due to fewer effects on umbilical arterial acidity, ephedrine is a frequent choice due to less alpha-adrenergic action on the uterine vasculature, which preserves uterine blood flow.[3] Also, it is sometimes given intramuscularly to achieve a longer-lasting effect.[4]

Since ephedrine results in both alpha and beta-adrenoreceptor stimulation, it is most beneficial in the setting of hypotension along with bradycardia or a low-normal heart rate, and caution is necessary for patients with hypotension and coexisting tachycardia. As with most vasopressors used during anesthesia, its use should be viewed as a temporizing measure while determining and addressing the source of hypotension. Another notable indication of ephedrine is for allergy relief. Due to constriction of smooth muscle via alpha receptor binding and bronchi dilation due to beta-2 receptor agonism, most sympathomimetics, including ephedrine, work effectively to decrease allergic symptoms.[5]

Oral formulations of ephedrine have been used historically to treat asthma via pulmonary vasoconstriction and reduction in airway edema along with beta-induced bronchodilation, but it is rarely used for this purpose in modern medicine due to unwanted cardiac effects and availability of more selective beta-agonists such as albuterol.[6]

Other less typical indications include bronchoconstriction, myasthenia gravis, and weight loss(in the past). In the early 2000s, the drug became very popular among over-the-counter supplements because it was found to work synergistically with caffeine to promote weight loss by stimulating metabolism.[7] However, ephedrine lost favor as a supplement after research established links to ventricular arrhythmias, nausea, and psychiatric issues with use.[8] Patients with myasthenia gravis may also benefit from treatment with ephedrine; however, evidence for its specific mechanism of action and benefit in this disease has not come from randomized controlled trials and only observational, and non-randomized trials are currently available.[9] Lastly, case reports demonstrate the use of ephedrine to alleviate bronchoconstriction during induction for patients on beta-blocker maintenance therapy.[10]

Mechanism of Action

Ephedrine, a stereoisomer to better-known pseudoephedrine, is a sympathomimetic amine with unique effects due to its indirect mechanism than other sympathomimetic agents like pseudoephedrine and phenylephrine.[11] Ephedrine acts as both a direct and indirect sympathomimetic. It binds directly to both alpha and beta receptors; however, its primary mode of action is indirectly achieved by inhibiting neuronal norepinephrine reuptake and displacing more norepinephrine from storage vesicles.[12] This action allows norepinephrine to be present in the synapse longer to bind postsynaptic alpha and beta receptors.[13] Ephedrine's indirect mechanism results in a sustained or even increased heart rate due to norepinephrine's ability to bind alpha and beta receptors, whereas more direct sympathomimetics like phenylephrine results in reflex bradycardia. While the indirect effect is most profound on arterial blood pressure, the direct vasoconstricting action functions more on the venous system. It is, therefore, effective in elevating central venous pressure when the patient is fluid challenged.[12]

Stimulation of alpha-1-adrenergic smooth muscle receptors within vasculature results in a rise in systemic vascular resistance and, consequently, systolic and diastolic blood pressure. Direct stimulation of beta-1-receptors by ephedrine and norepinephrine increases cardiac chronotropy and inotropy. Finally, beta-2-adrenergic receptor stimulation in the lungs results in bronchodilation with ephedrine administration, though it is not as pronounced as its cardiovascular effects.[14]

Pharmacokinetics

Ephedrine is metabolized into norephedrine. The parent drug and its metabolite are both excreted in the urine. Limited data after intravenous administration of ephedrine reinforce similar observations of urinary excretion. The plasma elimination half-life of ephedrine after oral administration is approximately 6 hours. Ephedrine can cross the placental barrier.[15]

Administration

Dosage Formulations

- Ephedrine sulfate oral tablet: 25 mg
- Ephedrine hydrochloride intravenous solution: 23.5 mg/5 mL
- Ephedrine sulfate intravenous solution: 5 mg/mL; 50 mg/mL

For treatment of asthma, 25 mg tablets are used every four hours as needed, and the maximum dose should not exceed 150 mg in 24 hours. For adults, bolus dose recommendations are at 5 to 10 mg, and intramuscular doses for a prolonged desired effect are at 25 to 50 mg.[12] Ephedrine requires dilution before intravenous use. In treating hypotension, cardiovascular effects via the indirect mechanism depend on adequate native norepinephrine stores. Tachyphylaxis will develop with prolonged and repeated use as this depletes endogenous norepinephrine stores. For this reason, it is administered in intravenous boluses when used intravenously and rarely as a continuous infusion.

Specific Patient Populations

Patients with Renal Impairment: According to the manufacturer's prescribing information, Ephedrine and its metabolite are excreted in the urine. In patients with kidney disease, excretion of ephedrine is affected with a corresponding increase in elimination half-life, which will lead to the slow elimination of ephedrine and consequently prolonged pharmacological effect and potential adverse reactions. Therefore, monitor patients with renal impairment carefully after the initial bolus dose for adverse events.

Patients with Hepatic Impairment: no dosage adjustments have been provided in the package insert.

Pregnancy Considerations: Though ephedrine is frequently used in obstetric patients just before and during delivery, there is insufficient data to support using the drug in earlier stages of pregnancy, and no animal reproductive studies have been performed, making it is a category C drug in pregnancy**.** Ephedrine should be given to a pregnant woman only if indicated.[16]

Breastfeeding Considerations: A single published case report indicates that ephedrine is present in human breast milk. Nonetheless, no information is available on the effects of ephedrine on the breastfed infant or the effects on milk production. Clinicians should consider the risk-benefit analysis for the mother's clinical requirement for ephedrine and adverse reactions on the breastfed child from ephedrine. Administration to the nursing women is not recommended as there is a higher-than-usual risk for infants according to the package insert.

Pediatric Populations: The FDA has not formally established safety and effectiveness in pediatric populations**.**[2]

Adverse Effects

Ephedrine may produce palpitations, headache, dizziness, nausea, vomiting, restlessness, and anxiety in the conscious patient. Ephedrine is also arrhythmogenic, and clinicians should be cautious when administering patients predisposed to arrhythmias or taking other arrhythmogenic medications, particularly digitalis. When used long-term, the catecholamine excess can result in contraction band necrosis of the myocardium, which predisposes the heart to ventricular arrhythmias.[17]

Another common effect of ephedrine is an alteration in the time until the onset and duration of action of other drugs. This effect is most notable during induction when giving ephedrine for a hypotensive patient before rocuronium.[18][19]

Additionally, ephedrine may be hepatotoxic, but there seems to be unclear evidence. This idea stems from case reports where ephedra species containing several compounds resulted in liver damage instead of a direct correlation.[20]

A less common effect experienced mostly by illicit users of ephedrine and its derivatives is the development of radiolucent urological ephedrine containing stones and paranoid schizophrenia.[21]

Contraindications

Ephedrine is contraindicated in a patient with acute hypertension or tachycardia. Ephedrine increases both chronotropy and inotropy and therefore increases myocardial oxygen demand, and its use requires caution in patients with ischemic heart disease or heart failure. Additionally, it should be avoided in situations where tachycardia would be undesirable, such as aortic stenosis. Alpha-adrenergic stimulation caused by ephedrine results in contraction of the smooth muscle at the base of the bladder, resulting in resistance to urine outflow, and caution is necessary for patients with urinary retention and prostatic hyperplasia.[14]

The use of ephedrine should be avoided or used with caution within 14 days of MAOI therapy due to excessive norepinephrine availability at the synapse, which could cause a hypertensive crisis through the indirect sympathomimetic effect of ephedrine.[22]

Norepinephrine and phenylephrine are other appropriate choices to maintain blood pressure post-spinal anesthesia. Norepinephrine was shown to have fewer episodes of hypotension in comparison to ephedrine, and phenylephrine showed less extensive effects on umbilical artery acidity.[23][24]

Monitoring

Although there are no specific drug monitoring requirements, variations in gene codons for receptor type can yield various sensitivity to ephedrine. Specifically, variations at codons 16 and 27 for the beta-2 adrenoreceptor have significant differences for ephedrine requirement to combat hypotension post-spinal anesthesia. Due to variation in responses and the fact that vasoactive medications can immediately affect blood pressure, heart rate, and pulse oximetry require close monitoring.[25]

Toxicity

According to the manufacturer's prescribing information, an overdose of ephedrine can cause a rapid rise in blood pressure. In the case of an overdose, careful monitoring of blood pressure is recommended. If blood pressure continues to rise to an unacceptable level, parenteral antihypertensive agents can be administered at the clinician's discretion.

Signs and Symptoms: The principal manifestation of ephedrine poisoning is hypertension and convulsions.

Treatment: If respirations are shallow or cyanosis is present, secure the airway and provide mechanical ventilation.

Antidote: For hypertension, 5 mg phentolamine mesylate diluted in saline may be administered slowly intravenously, or clinicians may give 100 mg orally. Convulsions may be controlled by benzodiazepines.

Enhancing Healthcare Team Outcomes

Though anesthetists are some of the only providers to administer ephedrine routinely, they do not do so within a vacuum. All healthcare team members, including doctors, nurses, pharmacists, and physician assistants providing care for that patient, should know its effects and different routes of administration. When given intravenously, drug effects are often of short duration (minutes), and tachyphylaxis is common. However, if given intramuscularly, the vasopressor effects typically remain for 60 to 90 minutes.[12] These effects are of particular importance on the labor and delivery ward, where patients often receive intramuscular injections of ephedrine following spinal block to attenuate the sympathectomy and nausea that frequently accompany spinal blockade. The effects of an IM injection may last much longer than the procedure itself, and nurses and technicians should not rely on hypotension as the primary indicator of postpartum hemorrhage as IM ephedrine may mask this sign.

When planning to use ephedrine, the pharmacist should verify the dose and that there are no significant drug-drug interactions and report these findings to the clinical team. Nursing must be mindful of the adverse effects of the drug and be prepared to inform the clinician regarding their observations. Nursing will also commonly be involved in administration, so they must collaborate on dosing and administration with the pharmacist. These examples show how all interprofessional team members can optimize ephedrine therapy when necessary while mitigating its adverse effects. [Level 5]